A Note on Positional Encodings Beyond Sequence Length

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Abstract

We analyse the behaviour of rotary positional encodings (RoPE) when the sequence length at inference exceeds the training context window.
We derive a closed-form bound on the position-dependent attention decay.

1 Introduction

Long-context language models require efficient positional encodings that extrapolate beyond the training sequence length. Rotary Position Embedding (RoPE; Su et al., 2022) encodes position as a rotation in the complex plane, providing relative position information without explicit position embeddings. However, RoPE's behaviour on out-of-distribution lengths is poorly understood.

2 Analysis

Let theta_i = 10000^{-2i/d} for i in {0,...,d/2-1}.
The attention between positions m and n depends on cos((m-n)*theta_i).
For |m-n| > L_train, the cosine oscillates rapidly, reducing effective attention.

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References

[1] Su et al. RoFormer: Enhanced transformer with rotary position embedding. 2022.
[2] Press et al. Train short, test long: Attention with linear biases. ICLR 2022.